

Vibe Shuffle: Relative Valence–Arousal Adaptation for Music Selection

Markus Baumann
Ludwig-Maximilians-Universität
München
Munich, Germany
markus.baumann@campus.lmu.de

Felix Hajuj
Ludwig-Maximilians-Universität
München
Munich, Germany

Miriam Janner
Ludwig-Maximilians-Universität
München
Munich, Germany

Abstract

Music recommenders model durable preferences, but track fit also depends on current context. We study a baseline-relative next-song policy in *Vibe Shuffle*. Fixed heuristics express available facial cues relative to a personal reference on the Valence axis and available heart rate and beat-to-beat variability relative to a cardiac reference on the Arousal axis. Measurements collected during a track form a trajectory whose mean defines one of four recommendation regions. The policy selects the nearest unplayed song in the corresponding Valence–Energy region of a fixed catalog. The current export schema records the target, signal source, region, distance, quality flags, and ratings while excluding raw sensor data. An exploratory crossover left the effects on mood fit and liking unresolved. Evaluation is limited to the fixed catalog; sensor validation, complete trial records, and a larger preregistered study are needed.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → *Recommender systems*.

Keywords

affective computing, music recommendation, relative affect, valence and arousal, heart-rate variability, facial cues, baseline-relative adaptation

1 Introduction

Music choice is contextual. A track that suits exercise may be unwelcome during a quiet break. People use music to regulate mood and activation [11, 25]. Most recommenders prioritize stable signals such as listening history, favorite artists, similar users, and item features. These signals characterize usual preference and capture momentary fit only indirectly, a known challenge in contextual music recommendation [12, 26].

Vibe Shuffle studies a baseline-relative policy for music selection. Camera observations and optional wearable signals are expressed as within-listener changes and used to choose the next song rather than to assign absolute emotion labels. Full affect recognition would require reliable links among facial behavior, physiology, context, and subjective experience. A camera captures visible movement. Cardiac timing also varies with respiration, posture, activity, medication, and fitness. Neither channel directly identifies emotion.

During setup, the listener supplies short personal references. Later facial and cardiac observations are expressed as changes from those references. The horizontal axis represents relative Valence evidence, and the vertical axis represents relative Arousal evidence. Each point falls into a Tense, Energetic, Melancholic, or Calm recommendation region. The names identify catalog regions.

The catalog uses corresponding Valence and Energy coordinates. Vibe Shuffle averages the recent listener trajectory, retains unplayed songs from the matching region, and chooses the nearest one. A Random condition draws from the same remaining catalog without using the trajectory. The adaptive rule is *relative signals* → *recent point* → *region* → *nearest available song*.

Research question. How large and uncertain are the differences in participant-rated mood fit between the baseline-relative policy and Random selection from the same fixed catalog? Liking is treated as a secondary outcome.

Research contributions.

- **Baseline-relative operationalization.** Personal reference periods center camera and cardiac observations before they enter the relative Valence and Arousal coordinates.
- **Specified selection policy.** Event-driven observations form a trajectory; its mean defines a region, and distance selects the nearest unplayed song from the matching region.
- **Paired evidence.** The 15-participant crossover estimates mood-fit and liking differences against Random selection from the same frozen catalog under masked condition labels and identical rating questions.

The crossover left both effects unresolved and does not establish performance in a live streaming catalog. A platform study would require validated sensor mappings, complete trial records, preregistration, and a dynamic-catalog protocol.

2 Background

2.1 A relative valence–arousal coordinate

The circumplex model represents affect along two continuous dimensions: *valence*, from unpleasant to pleasant, and *arousal*, from low to high activation [22, 24]. Thayer’s account of energetic and tense arousal offers a complementary vocabulary for changes in activation and mood [28]. Vibe Shuffle uses the Valence–Arousal axes and selected category labels from these models. Both dimensional and discrete models appear in music-emotion research, and results depend on the selected model and annotations [7].



This work is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.

Figure 1 juxtaposes the two affect vocabularies with group-level HRV-frequency evidence used to motivate an exploratory diagnostic signal.

The recommendation rule maps sensor-derived coordinates onto the two-axis recommendation space. It does not output the emotion terms or coherence profiles shown in Figure 1.

Each listener point (V, A) records change from personal reference measurements. The center marks the reference. More positive facial display moves the point right, and greater cardiac activation moves it upward. Facial movements vary by person and context and do not uniquely identify emotions [5]. Body context also shifts facial judgments, and Valence–Arousal relations vary across people and cultures [2, 14].

2.2 Connecting listener state to music

Songs can be placed in a parallel two-dimensional space. Song *Valence* describes conveyed positivity, whereas *Energy* describes perceived intensity and activity. Prior emotion-oriented music interfaces have used these attributes as musical counterparts to affective valence and arousal [9]. Vibe Shuffle compares listener Valence with song Valence and listener Arousal with song Energy. Musical Energy and physiological Arousal remain distinct constructs. Continuous music-emotion coordinates also depend on subjective annotations [30]. Music unfolds over time, so a single song coordinate is an average approximation of a potentially changing emotional course [1].

Splitting each axis at 0.5 yields four recommendation regions. The listener’s relative position selects a region. Quadrant membership determines eligibility, and Euclidean distance ranks the eligible songs.

2.3 Heart-rate variability as relative evidence

Heart-rate variability (HRV) describes variation between successive heart beats. The root mean square of successive differences (RMSSD) is a widely used time-domain summary of short-term RR-interval variation [27]. Heart rate and RMSSD depend on respiration, movement, posture, fitness, medication, sensor quality, and other factors. These measures lack emotion-specific interpretation. Recommended practice includes standardized conditions, artifact review, matched durations, and respiration assessment [15].

Vibe Shuffle uses cardiac measurements as baseline-relative evidence for the Arousal axis. Heart rate above the reference and RMSSD below it contribute upward evidence; changes in the opposite direction lower it. The cardiac rule has not undergone clinical or psychological validation.

3 Related Work

3.1 Emotion-aware music selection

Music can evoke, express, maintain, or help regulate affect through several mechanisms [11]. Emotion-aware music systems use listener-provided or inferred context to constrain the next track.

Helmholz et al.’s *Feel the Moosic* uses a two-axis interface. Its color-coded representation simplifies the Russell and Thayer models. A listener selected a position, and the prototype requested songs within corresponding Spotify Valence and Energy ranges [9]. Vibe Shuffle retains the two-axis mapping and derives the listener

coordinate from baseline-relative camera and cardiac evidence. It summarizes a trajectory over the listening window and selects an unplayed catalog track near the mean coordinate. The experimental interface reports the resulting target, region, and distance.

3.2 Personal and multimodal affective recommenders

Prior systems have used personal physiological or camera cues to select music. Janssen et al. learned listener-specific relations between music and physiological response and used them to steer an affective music player toward a goal [10]. Ayata et al. compared machine-learning models that predicted Valence and Arousal from wearable galvanic-skin-response and photoplethysmography measurements before recommending music [3]. Tran et al. used a facial image to estimate a point in Valence–Arousal space and retrieved nearby songs in Spotify feature space [29].

Vibe Shuffle applies a fixed rule that maps recent within-person changes to a catalog coordinate and evaluates participant/session-level recommendation ratings.

3.3 HRV and coherence evidence

HRV provides useful information about cardiac dynamics, but its interpretation depends on measurement duration and context [27]. Balaji et al. analyzed 1,884,216 mobile HRV-biofeedback sessions from 70,041 users and reported aggregate associations between coherence measures and emotional labels entered by users after their sessions [4]. Sessions labeled with positive emotions had higher mean coherence and more stable coherence frequencies than sessions labeled with negative emotions. The study was an observational analysis of 3–15 minute biofeedback practice, lacked pre-session emotion reports, and did not establish a per-window emotion classifier.

Vibe Shuffle logs a different custom spectral-peakedness proxy for diagnostic analysis only. Selection instead uses the baseline-relative heart-rate and RMSSD estimate.

3.4 Explanations and user experience

Recommender explanations can target transparency, trust, effectiveness, or decision support [21]. The Vibe study interface displays the target, region, eligibility status, and distance. Perceived quality, effort, privacy, and context remain separate aspects of user experience [13].

3.5 Comparison with prior systems

Vibe Shuffle differs from the cited systems in two study-design choices: a fixed baseline-relative mapping and a Random condition drawn from the same frozen catalog.

4 System Design and Selection Policy

4.1 System overview

Vibe Shuffle implements its research protocol in a client-side React/Vite browser interface. Camera input, an optional Bluetooth Heart Rate Service device, and Spotify playback are integrated within one experimental session. The current implementation has no study database or recommendation server. The selection policy

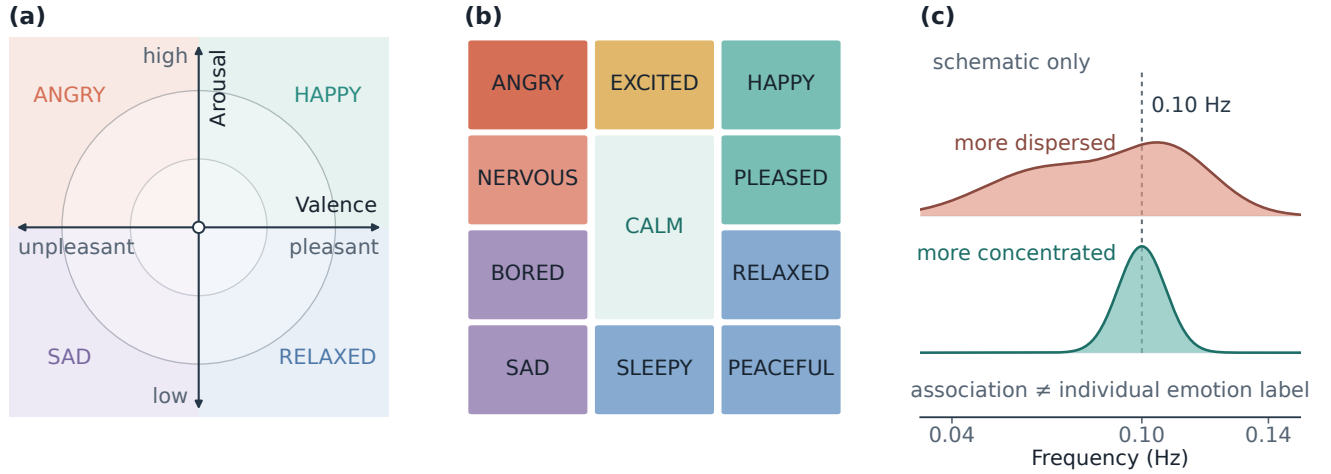


Figure 1: Conceptual foundations. Panels (a)–(b) are original redrawings adapted from Helmholtz et al. [9], who operationalized Russell’s circumplex [24] and Thayer’s mood–arousal model [28]. Panel (c) is an original schematic adapted from the group-level frequency distinction reported by Balaji et al. [4]; it contains no empirical observations.

updates between tracks: the latest listening window is summarized before the next track is selected. This interval-level update produces one decision per track and prevents brief fluctuations from changing playback immediately.

Figure 2 summarizes the process. Camera and cardiac measurements are expressed relative to personal reference values. During confirmed playback, successive coordinates form a trajectory. Its mean selects a region and a nearby song. The participant rates the track after playback.

4.2 Relative Valence from camera cues

MediaPipe Face Landmarker runs locally in the browser [17]. It returns facial blendshape activations such as smile, cheek, brow, eye, jaw, frown, and mouth movement. A fixed rule combines a selected subset into positive, negative, and tension-related scores. Vibe Shuffle defines these scores; they are absent from the MediaPipe output.

The interface schedules up to 14 seconds for a short facial reference, after which a slow running baseline continues to adapt. The participant can skip this step; completion time is not a separate export field. Later observations are debiased against the reference and smoothed across frames. Positive facial evidence moves Valence to the right; negative or tension-related evidence moves it to the left. If no face is detected, Valence returns to the neutral center. The current heuristic treats visible movement as additional positive-Valence and upward-Arousal evidence. Dancing, repositioning, lighting changes, and camera noise can produce the same signal. The movement rule lacks validation as a measure of engagement or affect.

4.3 Relative Arousal from cardiac timing

The optional wearable provides device-produced beats per minute and, when available, RR intervals—the elapsed time between successive beats. The research system receives device-produced summaries and has no access to a raw electrocardiogram or the device’s peak-detection process. Implausible and abruptly changing RR intervals are rejected before short-term variability is calculated.

The primary variability feature is the root mean square of successive differences (RMSSD). A 120-second reference stores the listener’s median heart rate and RMSSD together with robust estimates of their variation. During listening, higher-than-reference heart rate and lower-than-reference RMSSD provide upward Arousal evidence. Heart rate receives the larger influence because RMSSD estimated from a short window is unstable; when the two disagree, the RMSSD contribution is strongly reduced. At least 20 accepted RR intervals are required for a usable full estimate. Muñoz et al. found high agreement between 120-second and longer RMSSD recordings under controlled resting conditions [19]. Their study did not examine short wearable windows during movement or Arousal estimation.

The local study export also includes SDNN and a custom normalized frequency-domain spectral-peakedness score, recorded in the CSV as the physiology coherence field. This experimental diagnostic differs from the coherence measure analyzed by Balaji et al. and has no role in region selection. Balaji et al. report group-level associations for a different coherence measure [4]. Vibe Shuffle’s proxy has not been validated against that measure or against individual emotion labels. Song selection uses only the baseline-relative heart-rate and RMSSD estimate.

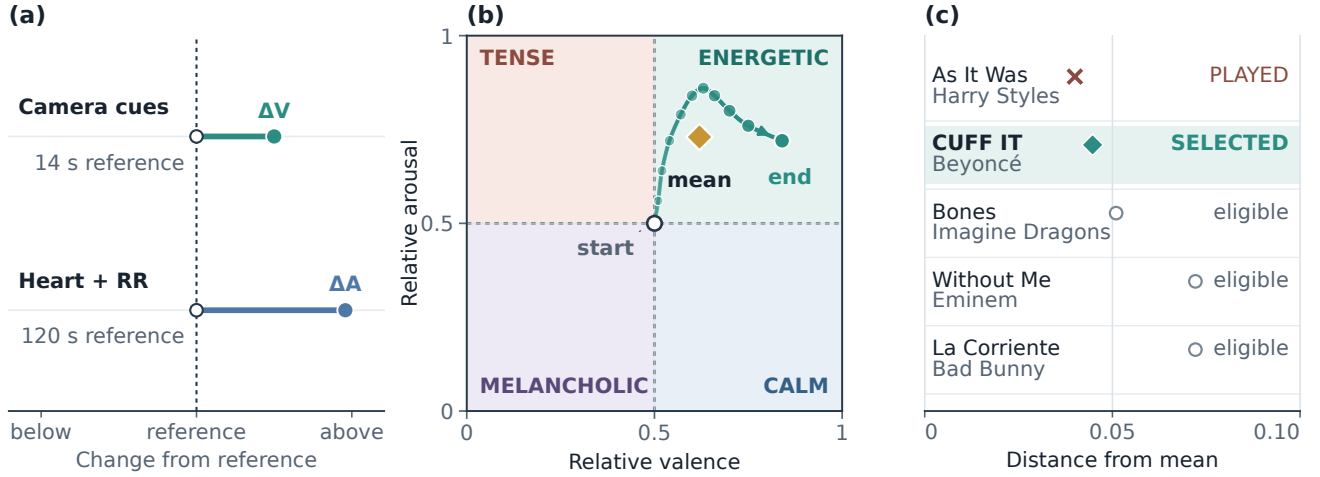


Figure 2: Baseline-relative sensing, trajectory aggregation, and catalog selection. (a) Camera and cardiac inputs are expressed as signed changes from personal references. (b) A curved Valence-Arousal trajectory is summarized by its arithmetic mean (gold diamond); points grow and darken from the open start marker to the filled endpoint. (c) Within the matching region, *As It Was* is nearest but excluded as played (cross), so *CUFF IT* becomes the selected eligible track (filled diamond); open circles are alternatives. Panels (a) and (b) are schematics. Panel (c) uses candidates and distances from the frozen catalog and reproduces the resulting selection. The trajectory illustrates the specified aggregation rule. The logged coherence field is excluded from selection.

4.4 Core fixed mapping

The following equations specify the part of the rule that produces the continuous coordinates. Let h , s , and t denote the baseline-debiased and smoothed happy, sad, and tense rule scores. The camera Valence coordinate is

$$V = \text{clip}_{[.05, .95]} \{ .5 + 1.05(h - .55s - .45t) \}.$$

Face samples arrive every 120 ms and use an exponential smoothing coefficient of .34. The continuing baseline coefficient is .002 and its debiasing factor is .9. Normalized movement m and nodding n , computed over 3.2 seconds, give camera Arousal

$$A_f = \text{clip}_{[.5, .95]} \{ .5 + .4m + .2n \}.$$

Movement and nodding also add $.38m + .42n$ to the happy evidence before smoothing. The archived file `src/expressionModel.js` lists the underlying blendshape gates and coefficients.

For cardiac input, z_H is the median-heart-rate change divided by its robust baseline scale, and z_R is the baseline-minus-current RMSSD change divided by its robust scale. The scale floors are 4 bpm and 6 ms. Define

$$d(z) = \begin{cases} 0, & |z| \leq .25, \\ \text{sign}(z) \min(|z| - .25, 3), & \text{otherwise.} \end{cases}$$

Cardiac Arousal is

$$A_c = \text{clip}_{[0, 1]} \{ .5 + .18[.75d(z_H) + w_R d(z_R)] \},$$

where $w_R = .35$, reduced to .07 when RMSSD indicates higher Arousal while heart-rate evidence is nonpositive. With usable cardiac data, $A = \text{clip}_{[0, 1]} \{ A_c + \max(0, A_f - .5) \}$; otherwise $A = A_f$.

Without a face, $V = .5$; without either usable channel, $(V, A) = (.5, .5)$.

4.5 Trajectory aggregation

Whenever the fused live coordinate changes during confirmed playback, Vibe Shuffle appends a point to the current trial buffer. Facial cues primarily control Valence. Usable cardiac evidence defines the Arousal base; visible movement can increase it. Without usable cardiac evidence, camera movement provides an upward-only fallback. If both channels are unavailable, the coordinate returns to $(0.5, 0.5)$.

The trial trajectory is the sequence

$$\mathcal{T} = \{(V_1, A_1), \dots, (V_m, A_m)\}.$$

At rating time, the selection policy uses the arithmetic mean

$$(\bar{V}, \bar{A}) = \frac{1}{m} \sum_{i=1}^m (V_i, A_i)$$

as the target for the next adaptive choice. Points are appended on coordinate changes, without a fixed sampling rate. The arithmetic mean is event-weighted. The export includes sample count and listening duration for review of this approximation.

4.6 Four regions and a fixed music space

Splitting both axes at 0.5 produces four recommendation regions: Energetic (higher Valence and Arousal), Calm (higher Valence and lower Arousal), Tense (lower Valence and higher Arousal), and Melancholic (lower on both axes). The frozen catalog contains 100 Spotify tracks, with 25 tracks in each corresponding Valence-Energy region. Track coordinates originate from historical Spotify features in a public dataset [18]. Energy serves as the catalog-side

proxy for Arousal. The audio feature and physiological construct are not equivalent. The repository preserves the 100 track records but not the dataset download date or a script that reconstructs their curation. The catalog is therefore reproducible as a frozen stimulus set, not as a fresh derivation from the source dataset.

Both Random and Vibe exclude the current and all previously played tracks. Random ranks the remaining tracks with a session-seeded pseudorandom score. Vibe first retains available tracks in the target region and then selects the smallest Euclidean distance:

$$s^* = \arg \min_{s \in Q(\bar{V}, \bar{A})} \sqrt{(V_s - \bar{V})^2 + (E_s - \bar{A})^2}.$$

The seed resolves only an exact distance tie. If a region has no remaining track, the policy falls back to the complete unplayed pool. The current export schema records the target coordinate, signal source, distance, and region match. The random seed, personal facial-reference values, and individual trajectory points are absent. The first track is ranked from the live setup estimate before the facial reference is finalized. An up-to-14-second overlay then blocks playback but may be skipped. Later choices use the preceding listening window.

4.7 Research data flow and local export

Spotify authentication uses Authorization Code with PKCE, and the Web Playback SDK provides playback for an allowlisted Premium account. Raw camera frames, facial landmarks, and Bluetooth notifications remain in browser memory. The locally downloaded CSV contains derived signal summaries, song metadata, selection fields, ratings, quality flags, pseudonymous identifiers, and timestamps. It is downloaded automatically at protocol completion and can also be downloaded manually. The file contains no images, landmark sets, raw waveforms, or audio. Its trajectory summaries allow review of the final target, but individual sensing steps cannot be replayed.

Spotify supplies playback for the frozen study catalog. The experiment does not evaluate a dynamically changing catalog or another streaming provider.

Authentication, asset delivery, and hosting require network requests to Spotify, MediaPipe, and GitHub Pages. Participant numbers and timestamps in the export are pseudonymous and should be handled as research data.

5 Exploratory Crossover Study

5.1 Protocol

The study compares the perceived appropriateness of the relative adaptive policy with Random selection from the same catalog. The protocol schedules five tracks in each condition. Two masked protocols reverse block order, and the study interface uses odd/even participant numbers to suggest an assignment. In participant-facing mode, the condition name is replaced by a protocol identifier. The experimenter can access the view toggle, protocol controls, and condition order in exported data. Masking depended on study administration and was not enforced by the study interface. The study was not double blind. Practice and carryover remain possible in the paired design; reversing block order helps control those effects [8].

Each track plays for up to 60 seconds. Participants may select *Rate now*; the actual duration and an early-rating flag are preserved.

Table 1: Exploratory condition summaries ($N = 15$ participants). Confidence intervals are two-sided. The reported study tests used one-sided Vibe-over-Random alternatives.

Outcome	Vibe M (SD)	Random M (SD)	Difference [95% CI]
Mood fit	4.71 (0.99)	4.15 (1.30)	+0.56 [−0.21, 1.33]
Liking	4.56 (1.18)	4.47 (0.77)	+0.09 [−0.52, 0.71]

After each track, the study interface collects a seven-point liking rating, a seven-point mood-fit rating, and a categorical current-mood report. For this exploratory analysis, mood fit is treated as the primary outcome because it measures the proposed match directly. Liking is secondary and separates contextual fit from general song preference. The prompts were *How much do you like this song?* (1, *Don't like it*; 7, *Love it*) and *How well did it fit your current mood?* (1, *Not at all*; 7, *Perfect fit*). The current-mood prompt offered Energetic, Calm, Tense, Melancholic, and Neutral in randomized order.

5.2 Available evidence

The supplied analysis report contains one five-trial condition mean per participant/session for 15 participants, together with paired tests and condition summaries. The archived analysis dataset contains 15 participant/session rows under generic analysis IDs. The protocol implies 75 planned ratings for each condition and outcome. Without the trial rows, completeness, exclusions, and within-session variation cannot be checked. Recruitment records, demographics, and the complete study-administration file are also unavailable. The available files limit the analysis to the 15 participant/session means. The human-study archive remains incomplete. In particular, the archive does not show how often a Vibe selection followed a camera-only, cardiac-only, fused, or no-signal path.

Paired differences are defined as Vibe minus Random. The analysis includes the supplied participant/session means, condition means and standard deviations, paired mean differences with two-sided 95% confidence intervals, and the originally supplied paired tests. No dated preregistration was available. We report the paired estimates, 95% confidence intervals, and supplied p -values as exploratory results [6, 20]; no confirmatory significance threshold was specified. The participant/session five-trial mean is the unit of analysis. Confidence intervals use the paired t distribution with 14 degrees of freedom. Figure 3 adds 20,000 paired bootstrap resamples with fixed seeds to show the distribution of the mean; its interval bars remain the t intervals, not bootstrap intervals.

The archived means are rounded to one decimal place. They reproduce the condition means, paired means, t statistics, and t intervals to rounding. Exact signed-rank results depend on the unavailable precision and tie convention, so the paper retains the supplied signed-rank values.

6 Results and Limitations

6.1 Mood-fit estimate and uncertainty

Mood fit averaged 4.71 under Vibe and 4.15 under Random (Table 1). The paired mean difference was +0.56 rating points with a 95% confidence interval from −0.21 to 1.33. The supplied directional

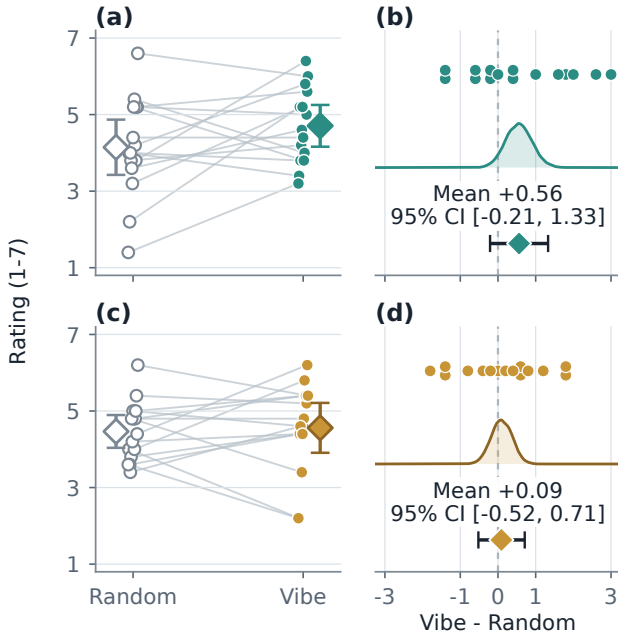


Figure 3: Participant-level paired outcomes. Panels (a)–(b) show mood fit; panels (c)–(d) show liking. Left panels connect Random and Vibe ratings and show condition means with 95% confidence intervals. Right panels show participant differences, bootstrap mean distributions, and paired mean 95% confidence intervals; both intervals cross zero.

paired test yielded $t(14) = 1.56$, $p = .070$; the signed-rank sensitivity check yielded $W^+ = 74.0$, $p = .088$. The standardized paired effect was $d_z = 0.40$. Eight participant/session means were higher under Vibe, one was tied, and six were lower. The 95% confidence interval includes effects on either side of zero. Figure 3 plots the paired participant means and Vibe–Random differences.

Liking averaged 4.56 under Vibe and 4.47 under Random. The paired difference was +0.09 with a 95% confidence interval from -0.52 to 0.71 , $t(14) = 0.32$, $p = .375$, and $W^+ = 58.5$, $p = .353$. The liking confidence interval includes zero; the study did not specify an equivalence margin.

6.2 Computational verification

The selection and signal-processing rules are pure JavaScript modules, allowing deterministic verification without camera, wearable, or Spotify access. The 63 tests cover catalog size and region balance, CSV finiteness and quality flags, facial-rule behavior, RR filtering, baseline-relative cardiac behavior, and Random/Vibe selection. All tests pass for the current reported source tree. They verify module-level computational behavior and contain no labeled affect data, independent sensor ground truth, or end-to-end browser, MediaPipe, Bluetooth, Spotify, and hardware checks. The archive does not identify the code revision used during the pilot sessions.

Both policies used the same playback path, rating questions, exposure count, and catalog. Their selected tracks could differ in artist, genre, language, familiarity, and popularity.

6.3 Measurement and study limits

The facial and cardiac mappings have not been validated against ground truth. Facial behavior cannot be reduced reliably to a universal readout of felt emotion [5]. Vibe’s rule weights were not learned or validated on a labeled affect dataset. Lighting, pose, occlusion, morphology, culture, and deliberate display can all change the output.

Heart rate and RMSSD also vary with respiration, posture, speaking, movement, fitness, medication, contact quality, and vendor processing [23, 27]. The listening window is short for HRV, respiration is not measured, and the research system receives processed RR intervals without access to raw peaks. Personal baselining adjusts between-person levels but leaves within-session physiological changes and sensor artifacts in the estimate.

Movement can affect the camera channel, raise heart rate, and introduce RR artifacts at the same time. One behavior can be counted by both channels. The trajectory mean weights coordinate-change events instead of elapsed time, and the first selection may occur before both references are fully established. Because sensing is optional, a Vibe selection can follow a camera-only, cardiac-only, fused, or no-signal path. The archived participant means do not report how often each path occurred.

At the exact no-signal center, current code paths disagree on which region owns the boundary: the fallback label is Calm, whereas the numerical greater-than-or-equal rule assigns $(0.5, 0.5)$ to Energetic. This edge case should be unified before a confirmatory study.

The study archive lacks raw trial rows, participant characteristics, recruitment and compensation details, the exact sensor model, room conditions, consent documentation, and the ethics procedure. The available archive therefore cannot establish recruitment, ethics, sensor setup, or trial-level data processing.

6.4 Materials and evidence availability

Source code, the frozen catalog, deterministic tests, figure scripts, and the 15 participant/session means are available at <https://github.com/eybmits/vibe-shuffle>. The JavaScript dependency versions are fixed in package-lock.json; the computational check is run with Node.js 20 using `npm ci` followed by `npm test`. A final submission should cite an immutable release of this archive.

The participant means and inferential summary were transcribed from the supplied reports; their source-file hashes are documented in `paper/RESULTS_PROVENANCE.md`. Trial-level ratings, sensor streams, administration records, and a code revision tied to pilot collection are not present. The current export schema and tests therefore describe the current implementation, not properties independently verified in the historical pilot sessions.

6.5 Next validation steps

Validation should begin with controlled changes in expression, lighting, posture, movement, and respiration applied to the frozen facial and cardiac mappings. A confirmatory study should then require successful calibration, timing, missing-data flags, and a complete ten-trial record. Before collecting a larger crossover sample, pre-register the meaningful mood-fit effect, sample size, exclusions, primary paired analysis, order analysis, and sensor-quality checks.

Base sample size on interval precision or a prespecified smallest relevant effect [16]; the pilot estimate is too uncertain to serve as the target.

Future protocols should specify whether the policy is intended to maintain, intensify, or shift the recent region. The outcome measure should match that stated goal.

A later study could evaluate the same policy on a larger, changing catalog. Its protocol should record playback failures, unavailable features, catalog changes, and recommendation coverage.

7 Conclusion

In the 15-participant crossover, the estimated Vibe–Random difference was +0.56 rating points for mood fit (95% CI [−0.21, 1.33]) and +0.09 points for liking (95% CI [−0.52, 0.71]). Both intervals include zero. Within the fixed catalog, the pilot leaves both effects unresolved and does not establish that baseline-relative selection improves what listeners hear next.

Vibe Shuffle supplies a specified policy that can be tested and audited. Personal references express camera and cardiac observations as relative Valence and Arousal evidence without assigning emotion labels. The trajectory-to-region-to-nearest-song rule turns those observations into a reviewable selection, and the paired means quantify direction and uncertainty against Random. The sensor mappings lack independent validation, while the historical archive lacks trial rows, signal-path frequencies, and a code revision tied to collection. Current tests cover the present source tree, not the historical sessions.

A confirmatory study should validate both mappings, resolve the no-signal boundary, compare camera-only, cardiac-only, fused, and Random conditions, evaluate time-weighted trajectory aggregation, preserve complete records, and preregister the target effect and analysis. A changing-catalog platform study can then test generalization. Vibe Shuffle makes the hypothesis precise and auditable; evidence that baseline-relative sensing improves what plays next remains to be produced.

References

- [1] Anna Aljanaki, Yi-Hsuan Yang, and Mohammad Soleymani. 2017. Developing a Benchmark for Emotional Analysis of Music. *PLOS ONE* 12, 3 (2017), e0173392. doi:10.1371/journal.pone.0173392
- [2] Hillel Aviezer, Yaacov Trope, and Alexander Todorov. 2012. Body Cues, Not Facial Expressions, Discriminate Between Intense Positive and Negative Emotions. *Science* 338, 6111 (2012), 1225–1229. doi:10.1126/science.1224313
- [3] Değer Ayata, Yusuf Yaslan, and Mustafa E. Kamasak. 2018. Emotion Based Music Recommendation System Using Wearable Physiological Sensors. *IEEE Transactions on Consumer Electronics* 64, 2 (2018), 196–203. doi:10.1109/TCE.2018.2844736
- [4] Sai Balaji, Nachum Plonka, Mike Atkinson, Malathy Muthu, Minvydas Ragulskis, Alfonsas Vainoras, and Rollin McCraty. 2025. Heart Rate Variability Biofeedback in a Global Study of the Most Common Coherence Frequencies and the Impact of Emotional States. *Scientific Reports* 15, 1 (2025), 3241. doi:10.1038/s41598-025-87729-7
- [5] Lisa Feldman Barrett, Ralph Adolphs, Stacy Marsella, Aleix M. Martinez, and Seth D. Pollak. 2019. Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements. *Psychological Science in the Public Interest* 20, 1 (2019), 1–68. doi:10.1177/1529100619832930
- [6] Geoff Cumming. 2014. The New Statistics: Why and How. *Psychological Science* 25, 1 (2014), 7–29. doi:10.1177/0956797613504966
- [7] Tuomas Eerola and Jonna K. Vuoskoski. 2013. A Review of Music and Emotion Studies: Approaches, Emotion Models, and Stimuli. *Music Perception* 30, 3 (2013), 307–340. doi:10.1525/mp.2012.30.3.307
- [8] Anthony G. Greenwald. 1976. Within-Subjects Designs: To Use or Not to Use? *Psychological Bulletin* 83, 2 (1976), 314–320. doi:10.1037/0033-2909.83.2.314
- [9] Patrick Helmholtz, Michael Meyer, and Susanne Robra-Bissantz. 2019. Feel the Moosic: Emotion-Based Music Selection and Recommendation. In *Proceedings of the 32nd Bled eConference: Humanizing Technology for a Sustainable Society*. University of Maribor Press, Maribor, Slovenia, 203–221. doi:10.18690/978-961-286-280-0.11
- [10] Joris H. Janssen, Egon L. van den Broek, and Joyce H. D. M. Westerink. 2012. Tune in to Your Emotions: A Robust Personalized Affective Music Player. *User Modeling and User-Adapted Interaction* 22 (2012), 255–279. doi:10.1007/s11257-011-9107-7
- [11] Patrik N. Juslin and Daniel Västfjäll. 2008. Emotional Responses to Music: The Need to Consider Underlying Mechanisms. *Behavioral and Brain Sciences* 31, 5 (2008), 559–575. doi:10.1017/S0140525X08005293
- [12] Marius Kaminskis and Francesco Ricci. 2012. Contextual Music Information Retrieval and Recommendation: State of the Art and Challenges. *Computer Science Review* 6, 2–3 (2012), 89–119. doi:10.1016/j.cosrev.2012.04.002
- [13] Bart P. Knijnenburg, Martijn C. Willemsen, Zeno Gantner, Hakan Soncu, and Chris Newell. 2012. Explaining the User Experience of Recommender Systems. *User Modeling and User-Adapted Interaction* 22, 4–5 (2012), 441–504. doi:10.1007/s11257-011-9118-4
- [14] Peter Kuppens, Francis Tuerlinckx, Michelle Yik, Peter Koval, Joachim Coosemans, Kevin J. Zeng, and James A. Russell. 2017. The Relation Between Valence and Arousal in Subjective Experience Varies With Personality and Culture. *Journal of Personality* 85, 4 (2017), 530–542. doi:10.1111/jopy.12258
- [15] Sylvain Laborde, Emma Mosley, and Julian F. Thayer. 2017. Heart Rate Variability and Cardiac Vagal Tone in Psychophysiological Research—Recommendations for Experiment Planning, Data Analysis, and Data Reporting. *Frontiers in Psychology* 8 (2017), 213. doi:10.3389/fpsyg.2017.00213
- [16] Daniel Lakens. 2022. Sample Size Justification. *Collabra: Psychology* 8, 1 (2022), 33267. doi:10.1525/collabra.33267
- [17] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. 2019. MediaPipe: A Framework for Building Perception Pipelines. doi:10.48550/arXiv.1906.08172 arXiv:1906.08172 [cs.DC]
- [18] maharshipandya. 2022. Spotify Tracks Dataset. Kaggle. <https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>.
- [19] M. Loretto Muñoz, Arie van Roon, Harriette Riese, Chris Thio, Emma Oostenbroek, Iris Westrik, Eco J. C. de Geus, Ron Gansevoort, Joop Lefrandt, Ilja M. Nolte, and Harold Snieder. 2015. Validity of (Ultra-)Short Recordings for Heart Rate Variability Measurements. *PLOS ONE* 10, 9 (2015), e0138921. doi:10.1371/journal.pone.0138921
- [20] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. 2018. The Preregistration Revolution. *Proceedings of the National Academy of Sciences* 115, 11 (2018), 2600–2606. doi:10.1073/pnas.1708274114
- [21] Ingrid Nunes and Dietmar Jannach. 2017. A Systematic Review and Taxonomy of Explanations in Decision Support and Recommender Systems. *User Modeling and User-Adapted Interaction* 27, 3–5 (2017), 393–444. doi:10.1007/s11257-017-9195-0
- [22] Jonathan Posner, James A. Russell, and Bradley S. Peterson. 2005. The Circumplex Model of Affect: An Integrative Approach to Affective Neuroscience, Cognitive Development, and Psychopathology. *Development and Psychopathology* 17, 3 (2005), 715–734. doi:10.1017/S0954579405050340
- [23] Karen S. Quigley, Peter J. Gianaros, Greg J. Norman, J. Richard Jennings, Gary G. Berntson, and Eco J. C. de Geus. 2024. Publication Guidelines for Human Heart Rate and Heart Rate Variability Studies in Psychophysiology—Part 1: Physiological Underpinnings and Foundations of Measurement. *Psychophysiology* 61, 9 (2024), e14604. doi:10.1111/psyp.14604
- [24] James A. Russell. 1980. A Circumplex Model of Affect. *Journal of Personality and Social Psychology* 39, 6 (1980), 1161–1178. doi:10.1037/h0077714
- [25] Thomas Schäfer, Peter Sedlmeier, Christine Städtler, and David Huron. 2013. The Psychological Functions of Music Listening. *Frontiers in Psychology* 4 (2013), 511. doi:10.3389/fpsyg.2013.00511
- [26] Markus Schedl, Hamed Zamani, Ching-Wei Chen, Yashar Deldjoo, and Mehdi Elahi. 2018. Current Challenges and Visions in Music Recommender Systems Research. *International Journal of Multimedia Information Retrieval* 7 (2018), 95–116. doi:10.1007/s13735-018-0154-2
- [27] Fred Shaffer and J. P. Ginsberg. 2017. An Overview of Heart Rate Variability Metrics and Norms. *Frontiers in Public Health* 5 (2017), 258. doi:10.3389/fpubh.2017.00258
- [28] Robert E. Thayer. 1991. *The Biopsychology of Mood and Arousal*. Oxford University Press, New York, NY.
- [29] Hieu Tran, Tuan Le, Anh Do, Tram Vu, Steven Bogaerts, and Brian Howard. 2023. Emotion-Aware Music Recommendation. *Proceedings of the AAAI Conference on Artificial Intelligence* 37, 13 (2023), 16087–16095. doi:10.1609/aaai.v37i13.26911
- [30] Yi-Hsuan Yang and Homer H. Chen. 2012. Machine Recognition of Music Emotion: A Review. *ACM Transactions on Intelligent Systems and Technology* 3, 3, Article 40 (2012), 30 pages. doi:10.1145/2168752.2168754